Project 2: MARSALA random-access MAC demonstration

CESE4055 Ad Hoc and Sensor Networks — Group Project 2

Objective and protocol

The project implements and demonstrates a challenging satellite random-access MAC receiver. A user places multiple replicas of one packet in randomly chosen slots of a frame. Slotted ALOHA accepts packets in singleton slots. CRDSA repeatedly decodes singleton packets and cancels their replicas (successive interference cancellation, SIC). MARSALA attempts to recover collided packets when SIC stalls: it correlates slot signals to localize replicas, aligns and combines them, then resumes SIC. The receiver changes; the transmission pattern is the same as CRDSA.

Simulator and live demonstration

The frame has 100 slots and 256 complex samples per slot. Each replica uses a random phase and a small amplitude variation; additive complex Gaussian noise has unit variance. The default replica SNR is -2.2 dB and the decoding threshold is a linear SINR of 0.61. The live GUI (`python3 live_demo.py`) has five controls: replica load, replica count, SNR, decoding threshold, and random seed. It updates the slot occupancy and decoded-packet comparison after slider changes, then animates the receiver's SIC and MARSALA trace. The New frame button advances the seed; Replay repeats the animation. The command-line trace (`python3 demo_frame.py --load 1.5 --replicas 2`) gives a reproducible example: 31 of 75 packets decode with ALOHA, 62 with CRDSA, and 75 with MARSALA. The 13 extra recoveries follow collision localization and combining.

Measurements and definitions

`python3 run_sweep.py --out results --frames 300` generates Monte Carlo plots and CSV data. Throughput is decoded packets per slot, packet loss is the undecoded fraction, and delay is the average number of frames until delivery among delivered packets. Pending packets retransmit in fresh slots each frame; they are dropped after 20 unsuccessful frames. The reported horizontal axis is **replica load**, `G = users * replicas / slots`, or transmissions per slot. Packet arrival load is `users / slots`; comparisons of different replica counts at the same G therefore include different numbers of users. The CSV retains `n_users` so equal-user comparisons can be made separately.

The throughput plot (`results/throughput_vs_load.png`) shows the effect of replica combining on packets delivered per slot. The loss plot (`results/loss_vs_load.png`) shows where collisions cease to be recoverable. The delay plot (`results/delay_vs_load.png`) shows how retransmissions add latency when a receiver stalls. The correlation-count plot (`results/marsala_correlations.png`) exposes the extra receiver work. These figures and their corresponding CSV files are the numerical evidence; throughput and loss points average

300 independent frames with a fixed random seed. Delay points come from 300 consecutive frames per persistent-user run.

Receiver	Peak throughput	Replica load at peak	Throughput at G=2.0	Loss at G=2.0
ALOHA	0.372	1.0	0.267	86.7%
CRDSA, R=2	0.538	1.3	0.368	63.2%
MARSALA, R=2	1.200	2.5	0.997	0.3%
CRDSA, R=3	0.675	2.1	0.662	1.2%
MARSALA, R=3	0.830	2.5	0.670	0.0%

At G=2.5, average delivered-packet delay is 4.62 frames for CRDSA R=2 versus 1.04 for MARSALA R=2; the corresponding R=3 values are 1.84 and 1.00 frames. These are conditional on the idealized receiver information.

Interpretation and limits

The clearest observation is the SIC deadlock in the sample frame: CRDSA leaves 13 packets unresolved, while combining lets MARSALA decode them and restart SIC. This also makes the receiver more computationally expensive. It examines other collided reference slots if the first candidate cannot be decoded, so correlation work can grow substantially at high load.

This is an **oracle-assisted concept simulation**. The decoder reads the generator's true occupancy, packet chip sequences, and amplitudes for decoding decisions. It does not infer all packet identities from received waveforms or perform physical interference cancellation. Its threshold is a stand-in for channel coding, and its delay model assumes immediate retransmission and no feedback overhead. The charts establish behavior of this idealized model, not measured throughput of a real satellite receiver. In particular, MARSALA R=2 reaches 1.20 packets per slot at G=2.5 by searching many reference slots (about 11,353 correlations per frame at that point). This large gain is a model artifact and must not be quoted as a real-world MARSALA performance prediction.

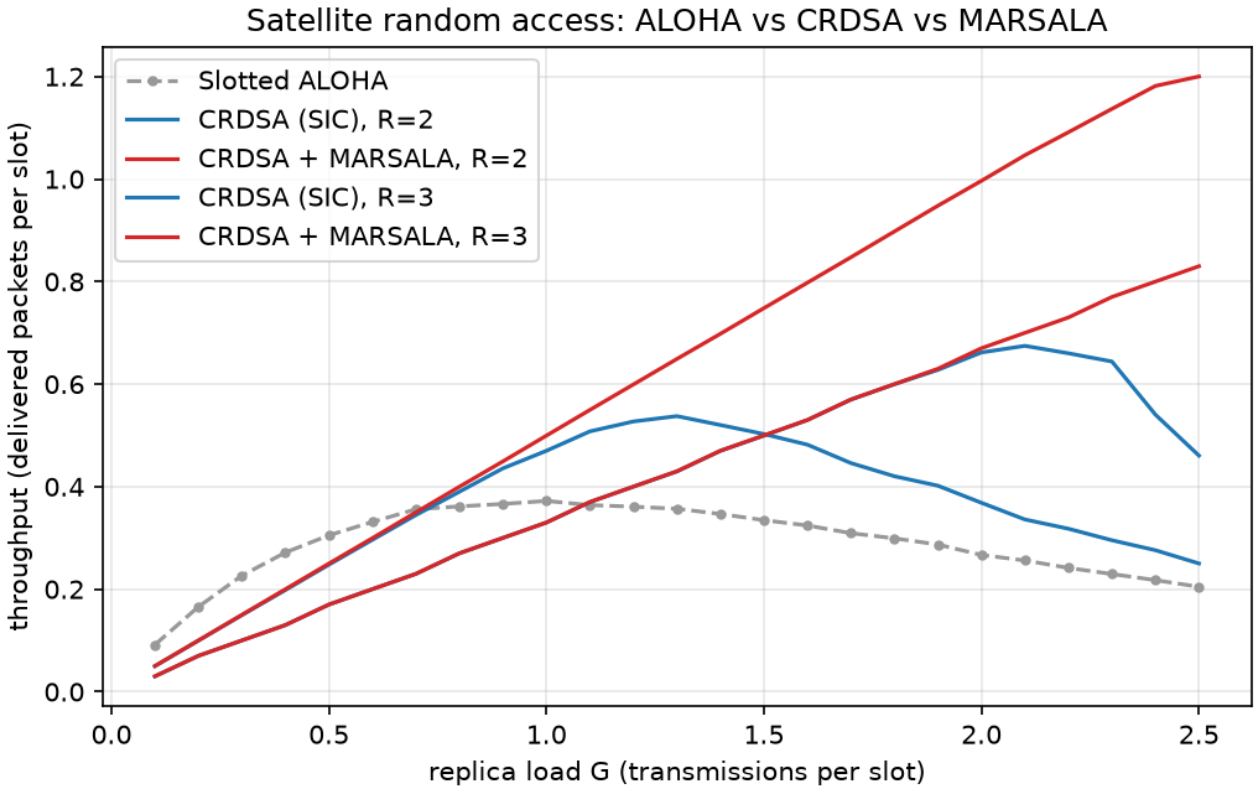
Reproduce

From `Project2/`, install `requirements.txt`, run `python3 live_demo.py` for the interactive demonstration, and run `python3 run_sweep.py --out results --frames 300` to regenerate the measurements. All sources and generated figures are in this directory.

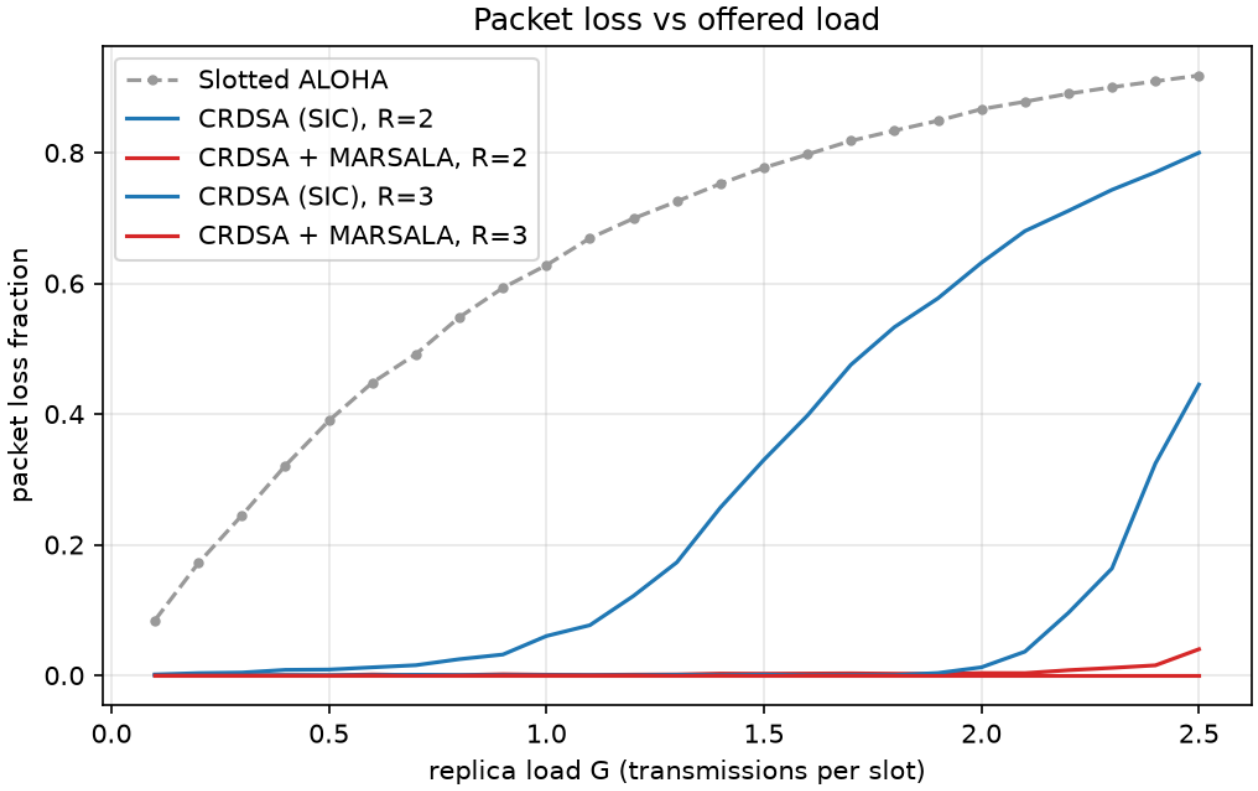
References

Zamoum, Lacan, Boucheret, Gineste, and Dupé, “A Multi-Replica Decoding Technique for Contention Resolution Diversity Slotted Aloha (MARSALA),” IEEE International Conference on Communications, 2015. CESE4055 Lecture 1, “Group Project: Project 2: MAC protocol,” slide 29.

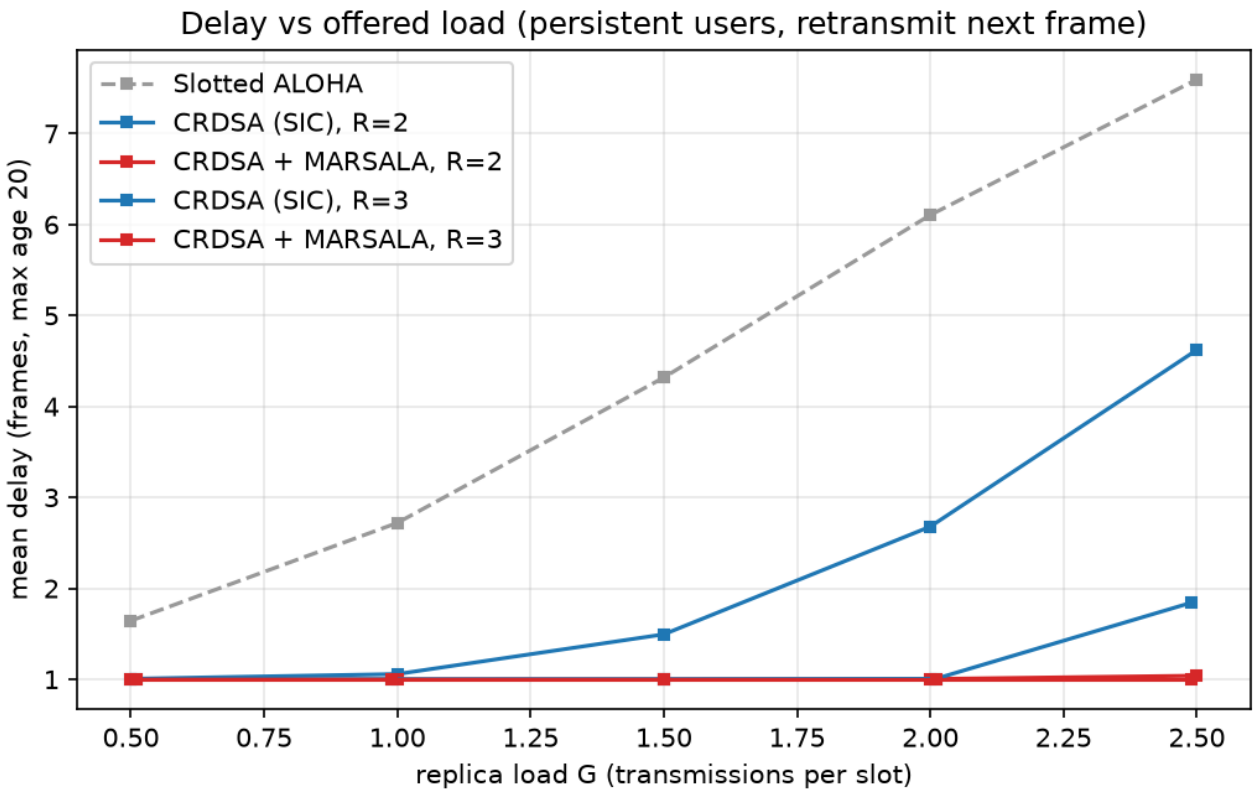
Results figures



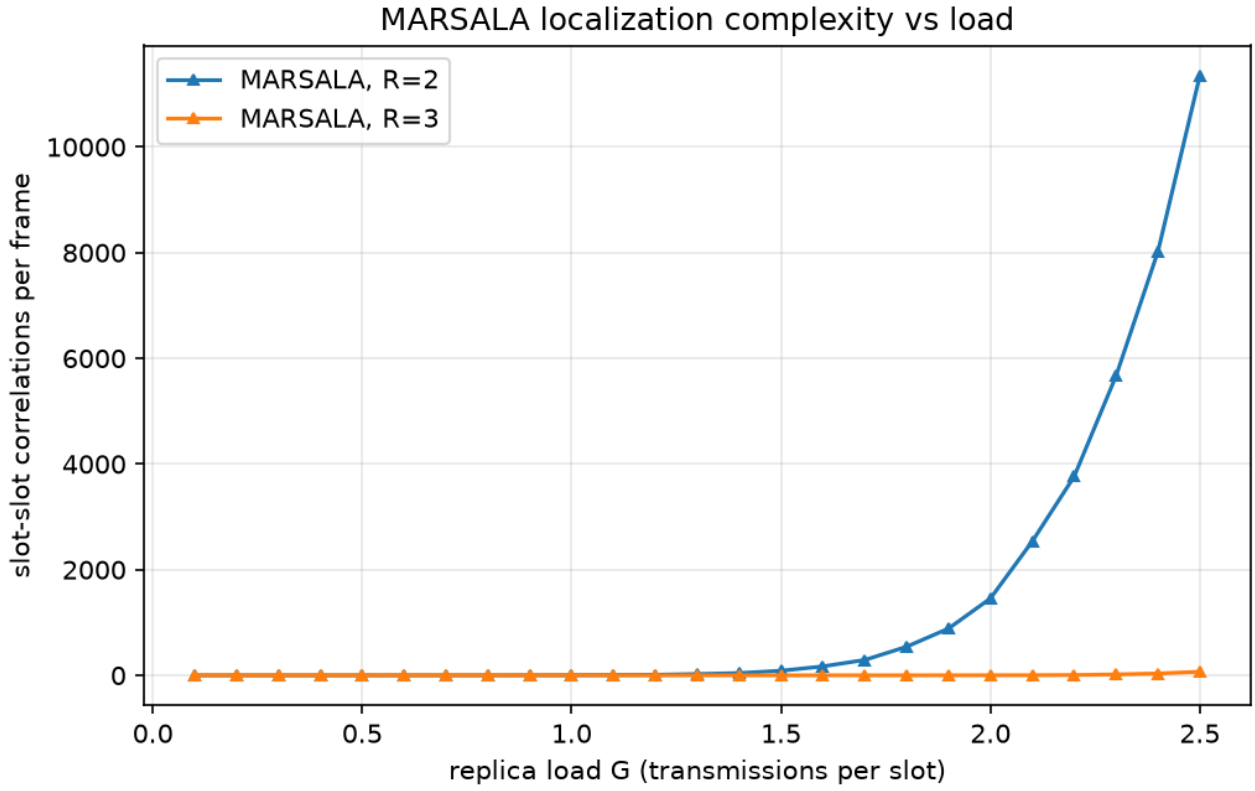
Throughput



Packet loss



Delay



Correlation work